

Streamlining Loan Approvals to Avoid Defaults

Using Loan Default Prediction To Save Time, Money, & Effort

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EXECUTIVE SUMMARY

- We currently lose \$20M for every 6K loans approved using the current manual process with a 20% default rate*
- Our ML model can accurately predict these loan defaults with an 81% recall preventing us from loaning out ~\$14+M of the \$20M
- The key indicators that predict a loan default (Debt-to-Income, Credit Delinquencies, Credit Age) can be explained**. Potentially we should make Debt-to-Income a required field***.
- The model will predict good loans with 74% precision which means it will reject some good customers resulting an opportunity cost loss of -\$630K.
- Automating the review process drastically reduces man-hours approximately saving \$594K****
- Given this, we recommend to pivot from a human review and start using an automated predictive process as soon as possible

** Based on 6K sample dataset which average of \$18.8K per loan*

*** Meets our regulatory obligations under the Equal Opportunity Credit Act*

**** When Debt-to-Inc Ratio is missing, 62% of those loans defaulted, 7x more than when it was present*

***** Assuming a cost reduction from \$100 to manually review a loan to \$1 using automation*



PROBLEM DEFINITION

Banks earn a considerable portion of their profits by collecting interest from retail loans

Lending money to customers who fail to repay loans cost bank significant profits and must be avoided.

Reliably deciding which loan applications banks should approve or reject is critical for banks to secure profits

Process to make a loan decision is manually intensive prone to human error and bias and can result in approving “bad loans” and rejecting “good loans”

PROBLEMS TO SOLVE

Can we reliably predict loan outcomes (will a loan default or be repaid) by using information on the application?

- What are the key indicators that determine loan outcomes?
- Can we minimize approving bad loans (avoid losing principal)?
- Can we minimize rejecting good loans (keep profit from interest)?
- Is the decision explainable?

Can we automate the loan approval process to reduce manual effort, improve consistency, and minimize bias?

- How should we implement this process in production?
- How can we ensure performance of the process over time?

PROPOSED SOLUTION

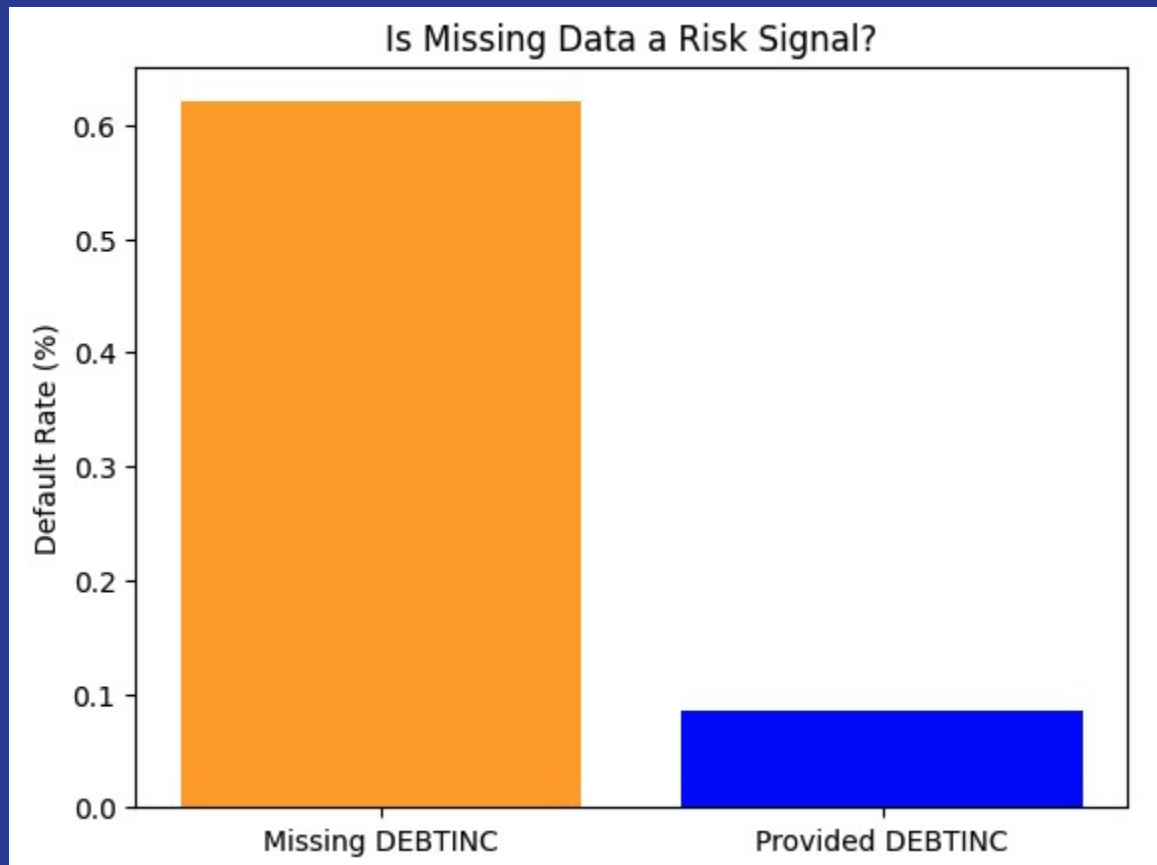
Automate using a binary classification model to accurately predict loan outcomes (specifically loan defaults) using a historical sample dataset

- Identify features which are the strongest predictors for a loan default from the input variables
- Optimize the model for minimizing false negatives (misclassifying a default) using a high recall metric and minimizing false positives (rejecting a good customer) using a high precision metric
- Make sure the predictions are explainable
- Address missing data or outliers
- Avoid "historical bias" in our model



EXPLORING THE DATA

- Data exploration from a historical sample of ~6K applications led to a few key takeaways*
- About 20% (1189) of loans approved defaulted resulting in a principal loss of ~\$20M
- Loans range in size from \$1.1K to \$89.9K, with a mean loan size of ~\$18.8K
- Some significant missing data (up to ~21%). Applicants that failed to provide their Debt-to-Income Ratio (DEBTINC) defaulted ~62% of the time vs ~9% for applicants that provided it.



MODELS TO CONSIDER

We will compare the performance of a several models (Logistic Regression, Decision Tree, Random Forest, and Gradient Boost)

A **Gradient Boosting** model is an ensemble "boosting" algorithm that builds trees sequentially, with each new tree specifically designed to correct the errors (residuals) of the previous ones.

- Aggressive at minimizing loss for complex datasets
- Capture non-linear relationships other models may miss
- Native support for missing values which our sample dataset has quite a few missing data columns.
- Less explainable (takes more effort to produce a reason)

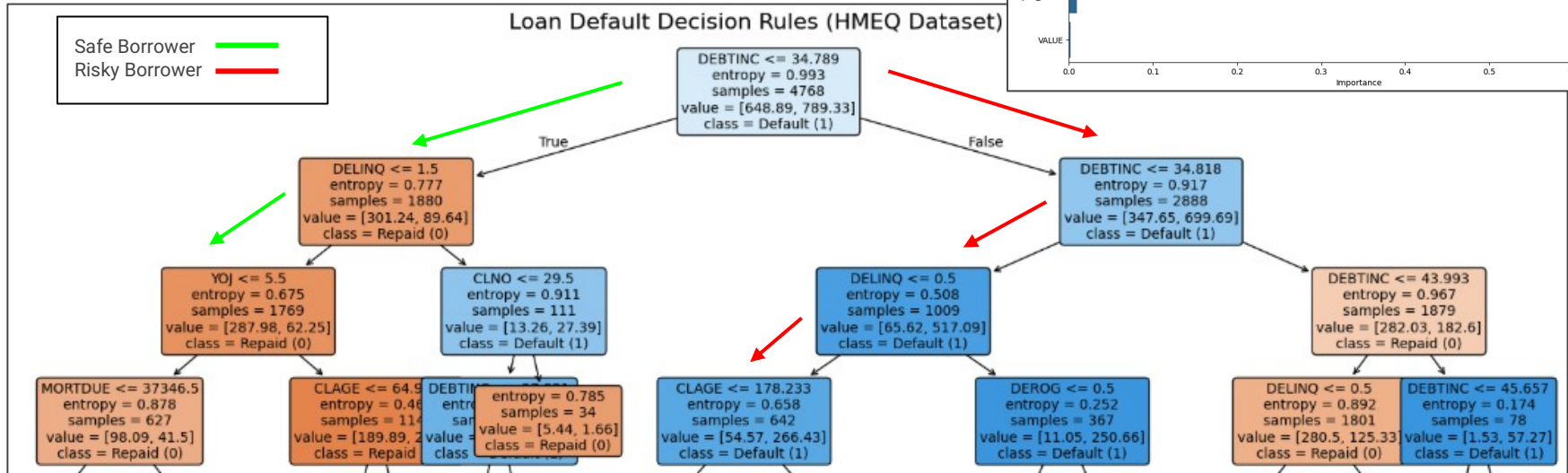
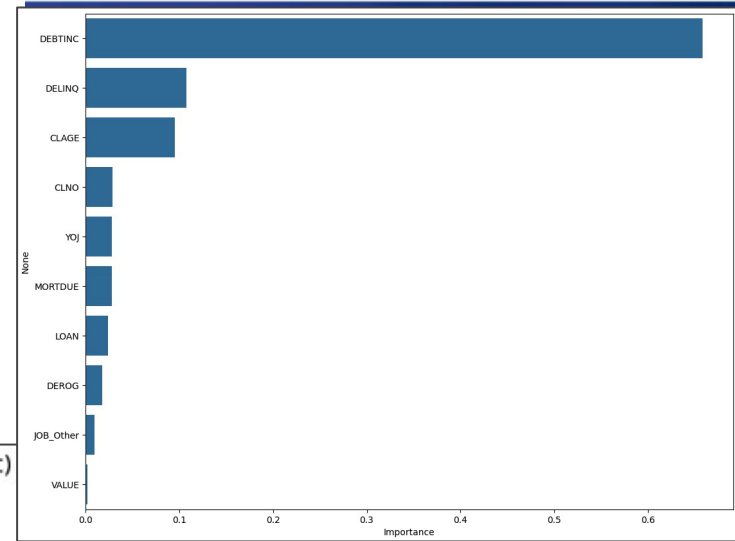
** See appendix for model definitions and comparison results.*



SELECTED MODEL

A tuned **Decision Tree** resulted in the best loan default recall metric of **81%** and **precision metric of 74%**

The key indicators that predict a loan default are Debt-to-Income (DEBTINC - 66%), Credit Delinquencies (DELINQ - 10%), and Credit Age (CLAGE - 9.5%).



* Considered Hyper Parameter Tuned Logistic Regression, Random Forest Models. See appendix for comparison.

HYBRID MODEL OPTION

A potential hybrid model solution: Ideally, if we could get the **Gradient Boost** tuning to match the performance of the **Decision Tree Recall** on bad loans we could use it in the final production model gains some precision and accuracy, and keep the **Decision Tree** visualization to explain the "logic" of the model to the bank's stakeholders.

Model	Treatments Applied	Precision	Recall / BAD	Accuracy	Comments
Decision Tree	-Tuned -Filled Missing Values* -Capped Outliers**	74%	82% / 81%	82%	Good Precision & Recall. 81% Recall for BAD loans Easier to explain
Gradient Boost	-Tuned -Native Missing value, Outlier treatment	78%	83% / 79%	85%	Good Precision & Recall. 79% Recall for BAD loans Need to use SHAP or LIME to translate the model's math into human language (e.g., "Denied due to high Debt-to-Income and recent delinquencies").

* Filled missing values using median imputation

** Removed outliers exceeding $1.5 * IQR$ (the spread of the middle 50% of a data set)



RISK & CHALLENGES

Risks

- Model Predictions are worse than expected in the real world
 - Data model was trained on is not real world representative (size, distribution)
 - Reduced too much predictive power in order to train (removing/capping outliers, filling in missing data, etc)
- Data drift and model decay could require you to re-examine and retrain the model
 - Changes over time to loan application fields (additional fields, removal of fields, changes to field semantics, scale, or precision)
 - Shifts in economic factors (impacting debt to income ratio thresholds, home values, etc.)
- Bank customer base shift
 - Banks starts to target higher income earners, bigger loans, and more risk

Challenges

- Implementing the automation in parallel with existing infrastructure
- Implementing monitoring on model performance
- Implementing proper change controls when updating model
- Conducting regular fairness audits and providing a rejection reasoning under the Equal Credit Opportunity Act



EXECUTING BUSINESS SOLUTION

Parallel Deployment

- Run the automated model in parallel with the existing manual approach for some time (~6 mos or target an application total)
- Compare both approaches and investigate any differences for model improvement
- Once comfortable with the automated model ("no worse than" if not demonstrably better performance) fully migrate over to it

Production Monitoring

- Monitor prediction accuracy over time and investigate if performance degrades
- Ensure proper change control on any to loan applications, model inputs, and model updates

THANK YOU

Appendix

SAMPLE DATASET VARIABLES

The Home Equity dataset (HMEQ) contains baseline and loan performance information for 5,960 recent home equity loans. The target (BAD) is a binary variable that indicates whether an applicant has ultimately defaulted or has been severely delinquent. This adverse outcome occurred in 1,189 cases (20 percent). 12 input variables were registered for each applicant.

BAD: 1 = Client defaulted on loan, 0 = loan repaid

LOAN: Amount of loan approved.

MORTDUE: Amount due on the existing mortgage.

VALUE: Current value of the property.

REASON: Reason for the loan request. (Homelmp = home improvement, DebtCon= debt consolidation which means taking out a new loan to pay off other liabilities and consumer debts)

JOB: The type of job that loan applicant has such as manager, self, etc.

YOJ: Years at present job.

DEROG: Number of major derogatory reports (which indicates a serious delinquency or late payments).

DELINQ: Number of delinquent credit lines (a line of credit becomes delinquent when a borrower does not make the minimum required payments 30 to 60 days past the day on which the payments were due).

CLAGE: Age of the oldest credit line in months.

NINQ: Number of recent credit inquiries.

CLNO: Number of existing credit lines.

DEBTINC: Debt-to-income ratio (all your monthly debt payments divided by your gross monthly income. This number is one way lenders measure your ability to manage the monthly payments to repay the money you plan to borrow.

OTHER EDA OBSERVATIONS

- The HMEQ dataset has significant missing values, particularly in DEBTINC (~21%) and DEROG/DELINQ (~10-12%). Handling these is critical because machine learning models cannot process NaN values.
- Home Improvement loans have a higher default rate (22.2%) compared to Debt Consolidation (18.9%) suggests that those borrowing to improve their homes are actually defaulting more frequently.*
- The raw value of the home itself is a very weak predictor of whether someone will default. A person with a \$1M home isn't necessarily more reliable than someone with a \$100k home in this specific pool of borrowers.
- There is a clear, non-random difference in the loan amounts requested by those who default versus those who repay. Bad loans tend to be smaller amounts requested by people with much lower property values and higher existing debt**
- The amount currently owed on an existing mortgage is a strong indicator of future default status. Borrowers who repay typically have more home equity (higher property value relative to their mortgage)**

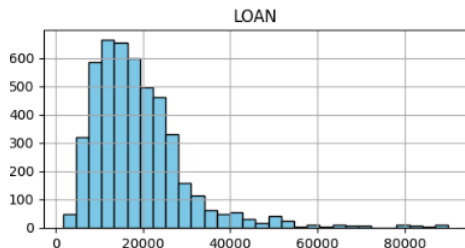
* Correlation analysis between BAD loans and REASON

** Using T-Score analysis. A T-score is a ratio that measures the "signal" relative to the "noise."

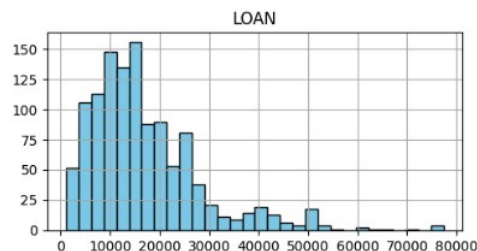


GOOD LOANS vs. BAD LOANS STATS

GOOD LOANS



BAD LOANS



	count	mean	std	min	25%	50%	75%	max
BAD	4771.0	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
LOAN	4771.0	19028.107315	11115.758554	1700.000000	11700.000000	16900.000000	23500.000000	89900.000000
MORTDUE	4359.0	74829.249055	43584.993587	2619.000000	47484.000000	66839.000000	93068.000000	371003.000000
VALUE	4764.0	102595.921018	52748.392952	8000.000000	67297.750000	90659.000000	120615.500000	471827.000000
YOJ	4321.0	9.154941	7.676033	0.000000	3.000000	7.000000	13.000000	36.000000
DEROG	4150.0	0.134217	0.514490	0.000000	0.000000	0.000000	0.000000	6.000000
DELINQ	4263.0	0.245133	0.674124	0.000000	0.000000	0.000000	0.000000	5.000000
CLAGE	4541.0	187.002355	84.465217	0.486711	120.219885	180.415787	240.157802	649.747104
NINQ	4336.0	1.032749	1.531322	0.000000	0.000000	1.000000	2.000000	11.000000
CLNO	4602.0	21.317036	9.682601	0.000000	15.000000	20.000000	26.000000	56.000000
DEBTINC	4290.0	33.253129	6.947482	0.720295	28.905127	34.541671	38.739077	45.569843

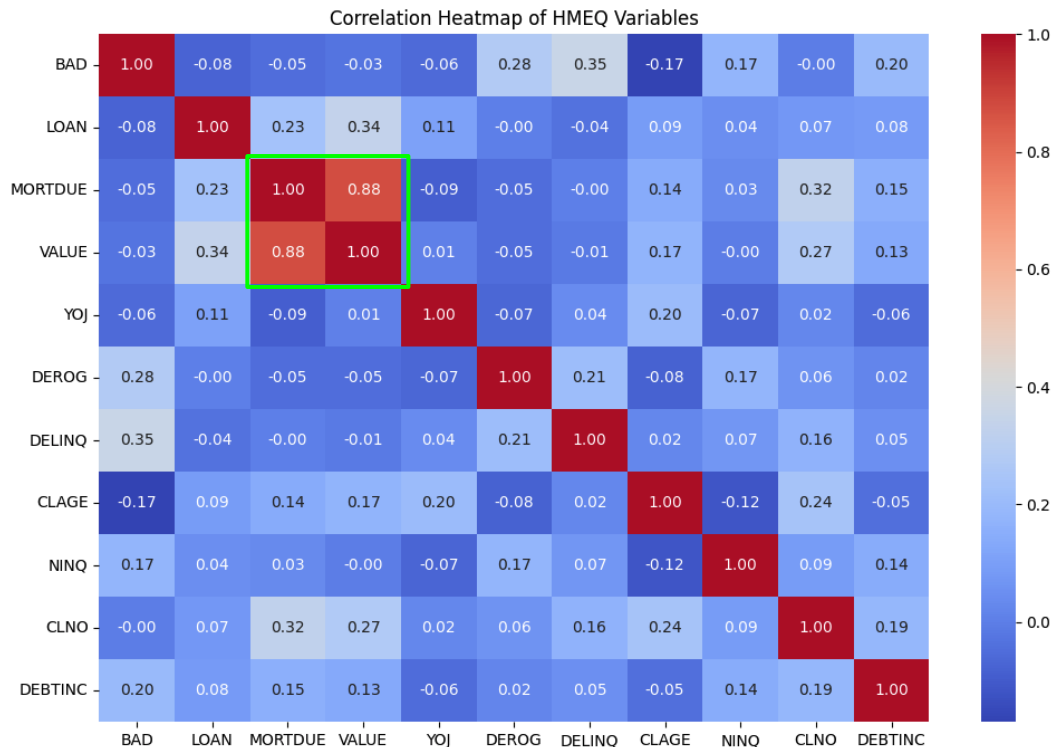
	count	mean	std	min	25%	50%	75%	max
BAD	1189.0	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.000000
LOAN	1189.0	16922.119428	11418.455152	1100.000000	9200.000000	14900.000000	21700.000000	77400.000000
MORTDUE	1083.0	69460.452973	47588.194467	2063.000000	39946.500000	60279.000000	85864.500000	399550.000000
VALUE	1084.0	98172.846227	74339.822506	8800.000000	59368.250000	82000.000000	116000.000000	855909.000000
YOJ	1124.0	8.027802	7.100735	0.000000	2.000000	6.000000	12.000000	41.000000
DEROG	1102.0	0.707804	1.468381	0.000000	0.000000	0.000000	1.000000	10.000000
DELINQ	1117.0	1.229185	1.902961	0.000000	0.000000	0.000000	2.000000	15.000000
CLAGE	1111.0	150.190183	84.952286	0.000000	96.033333	132.866667	193.283333	1168.233561
NINQ	1114.0	1.782765	2.246976	0.000000	0.000000	1.000000	3.000000	17.000000
CLNO	1136.0	21.211268	11.812981	0.000000	13.000000	20.000000	28.000000	71.000000
DEBTINC	403.0	39.387645	17.723586	0.524499	32.383046	38.079762	43.285990	203.312149

LOAN amount for Good loans (\$19,028) is actually HIGHER than for Bad loans (\$16,922). Bad loans tend to be smaller amounts requested by people with much lower property values and higher existing debt. A \$15k loan is "safer" for someone with a \$100k home than a \$10k loan is for someone already underwater.

MORTDUE amount for Good loans (\$74,829) is actually HIGHER than for Bad loans (\$69,460). Even though the dollar difference is only about \$5,300, the consistency of this gap across thousands of loans is what makes the p-value so small and statistically significant. Borrowers who repay typically have more home equity (higher property value relative to their mortgage). Borrowers with lower MORTDUE values in this dataset also tend to have lower property values (\$98k for defaults vs \$102k for repaid), meaning they have less of a financial "cushion" if they hit hard times.

CORRELATION MATRIX

This identifies if any variables are redundant (multicollinearity). For example, MORTDUE and VALUE are highly correlated.



SELECTING A MODEL

Model	Brief Definition	Pros	Cons	Suitability
Logistic Regression	Predicts the probability of a binary outcome (Yes/No) using a linear equation.	Simple to explain; fast; provides clear "odds ratios" for features.	Struggles with non-linear data; assumes features don't interact.	Small to Medium. Best for simple binary classification where "why" matters.
Decision Tree	A flowchart-like structure that splits data based on "if-then" rules.	Easy to visualize; handles missing data; no scaling needed.	Highly prone to overfitting; small data changes can flip the whole tree.	Small to Medium. Great for finding "smoking gun" rules (like your DEBTINC split).
Random Forest	An "ensemble" of many decision trees that vote on the final outcome.	Very stable; hard to overfit; excellent at handling "noisy" data.	Can be "slow" to run; hard to visualize (it's a "Black Box").	Medium to Large. The "Goldilocks" model for most tabular datasets.
Gradient Boost	Trees are built one at a time, each new tree "fixing" the errors of the last.	Most powerful/accurate; finds complex, subtle patterns.	Requires heavy tuning; sensitive to outliers and overfitting.	Large. Best for squeezing every bit of accuracy out of complex data.

MODEL COMPARISON

Model	Treatments Applied	Precision	Recall / BAD	Accuracy	Comments
Logistic Regression	-Scaling* -Tuned -Filled Missing Values** -Capped Outliers***	40%	50% / 61%	80%	Poor Precision & Recall. 61% Recall for BAD loans
Decision Tree	-Tuned -Filled Missing Values** -Capped Outliers***	74%	82% / 81%	82%	Good Precision & Recall. 81% Recall for BAD loans Easier to explain
Random Forest I	-Tuned -Filled Missing Values** -Capped Outliers***	81%	79% / 65%	88%	Good Precision & Recall. 65% Recall for BAD loans
Random Forest II	-Tuned -Removed Missing data rows (3364)	98%	82% / 63%	97%	-Trained on only fully populated data (20% data removed) -Could be used as a "Fast Lane" approval for applicants who fill out every field and clear the thresholds for the top 4 features then the bank can safely approve them instantly with a 98% precision rate.
Gradient Boosting	-Tuned -Native Missing Value, Outlier treatment	78%	83% / 79%	85%	Good Precision & Recall. 79% Recall for BAD loans Need to use SHAP or LIME to translate the model's math into human language (e.g., "Denied due to high Debt-to-Income and recent delinquencies").

* Scaling is required to normalize values in stddev across different features (EX: LOAN range in thousands, DELINQ in single digits)

** Filled missing values using median imputation



HYPERPARAMETER TUNING

While a model "learns" the data on its own, it can't decide its own rules—like how deep a tree should grow or how fast it should learn. If the settings are too loose, the model is under-trained; too tight, and it just memorizes the training data (overfitting). Tuning finds the "Sweet Spot" where the model performs best on data it has never seen before.

Model	Key Parameters to Tune	How Tuning Works	Tuned Parameters (using GridSearch*)
Logistic Regression	<code>C</code> (Regularization strength), <code>penalty</code> (L1/L2)	Regularization strength adjusts a "penalty" on the model's coefficients. It prevents any single feature (like DEBTINC) from having an unfairly massive influence.	[Didn't Tune - N/A]
Decision Tree	<code>max_depth</code> , <code>min_samples_leaf</code> , <code>criterion</code> (Gini/Entropy)	Tuning "prunes" the tree. It stops it from growing too deep and creating rules that only apply to a handful of specific people.	<code>DecisionTreeClassifier?</code> <code>idDecisionTreeClassifier(class_weight={0: 0.17, 1: 0.83}, max_depth=np.int64(6), min_samples_leaf=20, random_state=1)</code>
Random Forest	<code>n_estimators</code> , <code>max_features</code> , <code>max_depth</code>	It finds the right balance between the number of trees and how much "randomness" to allow so the trees don't all look the same.	<code>RandomForestClassifier(class_weight={0: 0.17, 1: 0.83}, max_features=0.7, min_samples_leaf=np.int64(3), n_estimators=500, random_state=1)</code>
HistGradient Boost	<code>learning_rate</code> , <code>max_iter</code> , <code>max_leaf_nodes</code> , <code>l2_regularization</code>	Tuning controls the speed and caution of the model. It ensures each new tree fixes an error without "over-correcting" too aggressively.	<code>HistGradientBoostingClassifier(class_weight='balanced', l2_regularization=1.0, learning_rate=0.01, max_depth=3, random_state=42)</code>

*GridSearchCV: You give a list of options (e.g., `max_depth: [3, 5, 10]`). It tries every single possible combination and tells you which one won. It's thorough but can be very slow.

CONFUSION MATRIX

Important: You can see individual class performance (BAD=1) which represents the loans we want to avoid. Our model achieve 81% Recall for this class and 82% overall.

Metric	Calculation	What it means for your Loans
Accuracy	$\frac{TP+TN}{Total}$	Overall, how often is the model right? (Can be misleading if most loans are good).
Precision	$\frac{TP}{TP+FP}$	When the model says "Default," how often is it actually a default? (Avoids annoying good customers).
Recall	$\frac{TP}{TP+FN}$	What % of all actual defaults did the model catch? (This is the most important for saving money).
F1-Score	$\frac{2 \times Prec \times Rec}{Prec + Rec}$	The "Balance." A high score means the model is good at both catching risk and keeping customers.

Decision Tree Model Results

...		precision	recall	f1-score	support
	0	0.94	0.82	0.88	954
	1	0.53	0.81	0.64	238
	accuracy			0.82	1192
	macro avg	0.74	0.82	0.76	1192
	weighted avg	0.86	0.82	0.83	1192

